Project: UART Communication on Vivado

Overview: UART (Universal Asynchronous Receiver Transmitter) is a widely used communication protocol in RS232 applications as well as in microcontrollers interfacing with sensors. UART is an asynchronous communication protocol, meaning it does not share a clock signal between the master and the slave. Instead, the two devices agree on a set baud rate that they will communicate on, and then data will be sent and received by both.

Specification:

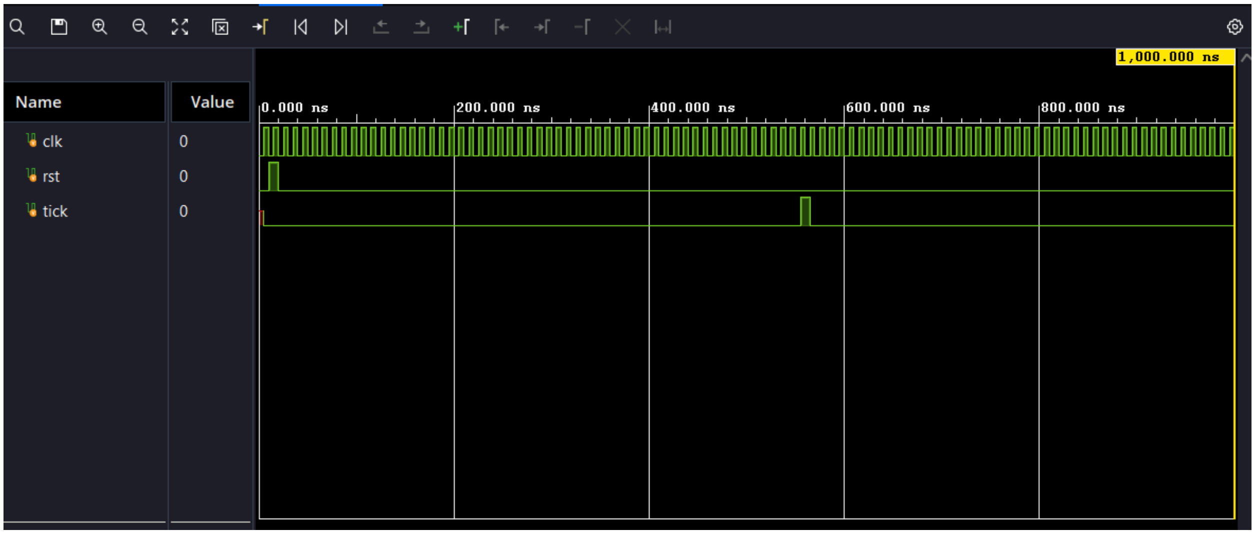
Baud Rate	9600, 115200
Data Bits	8
Parity	?
Start Bit	1
Stop Bit	1
Clock Frequency	?
Oversampling rate	?

System Architecture:

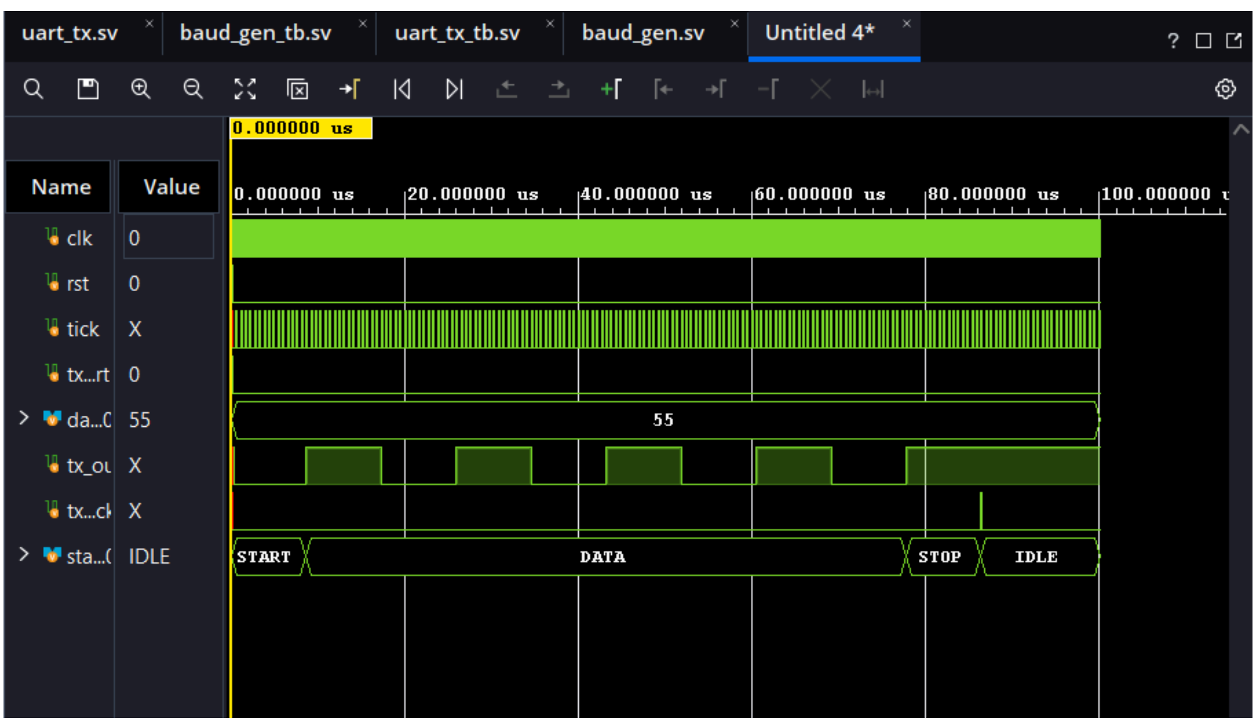
- **Modules**
    - Uart_tx
    - Uart_rx
    - Baud_gen

Baud_Gen File

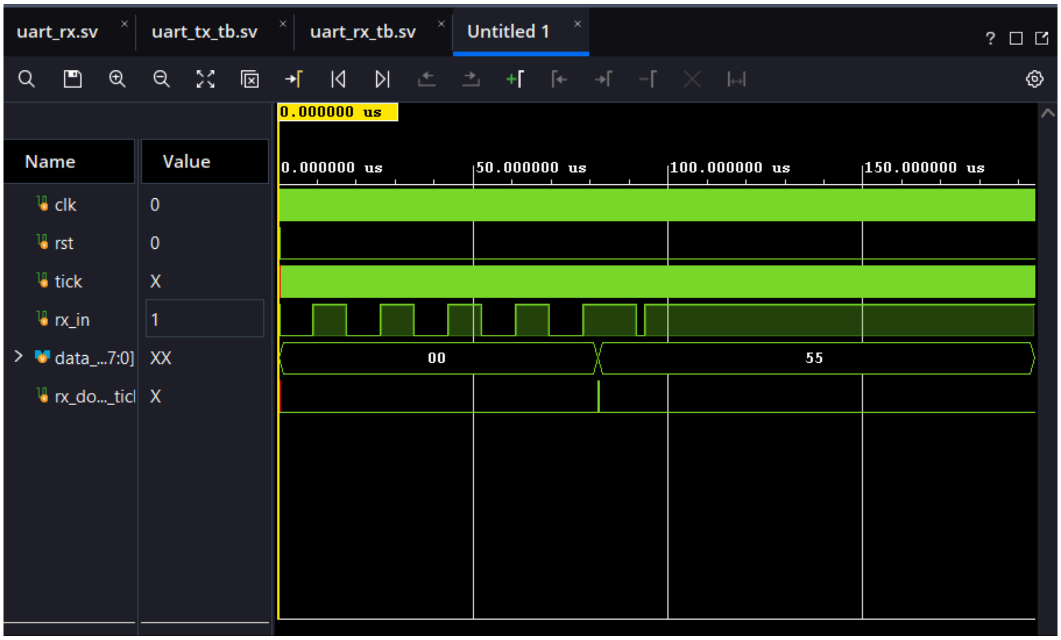
Testbench

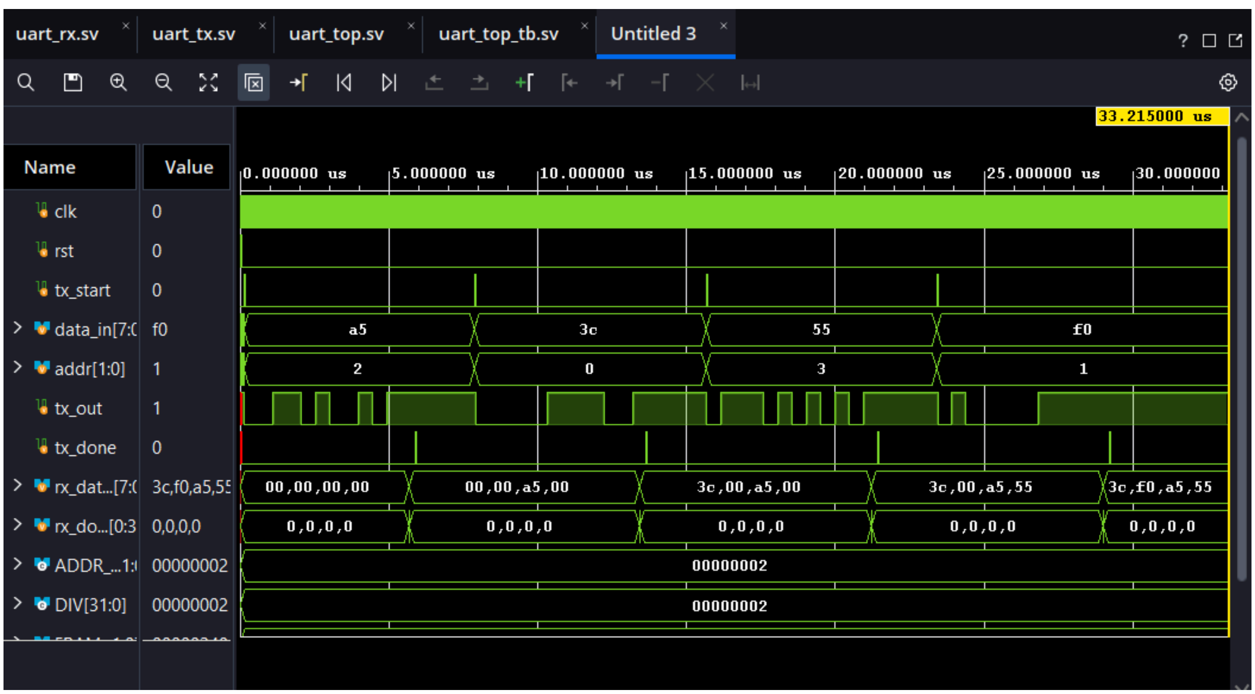


Uart_TX Testbench



Uart_RX testbench





One transmitter, four address-selected receivers on a shared line, with each byte being captured only by the addressed receiver